AirView™ EXCHANGE

Data exchange platform

AVX Patient Management Best Practices Guide

English

Contents

1 Overview

AirView Exchange (AVX) (formerly known as ResMed EasyCare Exchange/ECX2) is a complete integration Application Programming Interface (API) suite designed for two-way communication from certified applications into the ResMed AirView system.

This document covers best practices for the optimization of patient management APIs so you can design your integration with efficiency in mind. We highly recommend that you review each workflow to see what best practices you can apply to your integration setup.

2 Authentication

Before you continue, ensure your integrating application is correctly authenticated with AVX. You must include the authentication credentials for each call to AVX. For more information, refer to the *REST web service authentication* section in the *AVX API Technical Specification for Patient Management*.

3 Integrator types

AVX supports multiple integrator types. Most accounts are set up with one of the following integrator types:

Organization Specific Integrator: You have access to all patients in an HME's AirView organization. An HME administrator must provide access to your integrator account in AirView.

Patient Level Integrator: You only have access to patients that the HME has granted you permission to access. An HME clinical user can do this while they set up a new patient in AirView or edit an existing patient in their organization.

4 Match patients

4.1 Workflow overview

The instructions below outline how you can set up a patient matching framework to associate patients in your system with patients in AirView.

Follow this workflow when you set up your integrating application. You need:

- a patient matching framework to associate your system's patient ID with AirView's patient ID
- to reconcile your patient list in AirView if you did not create patients with your integrating application through AVX
- to have your integration recognize which ECNs were already processed.

Note: If you have pre-existing patients in AirView, but you want to manage the creation of AirView patients going forward, you need to complete this workflow.

4.2 Workflow tips

Your application must use **Get Accessible Organizations** regardless of what kind of integration it is. You can have multiple organizations assigned to each integrator type.

4.3 Required APIs

- **Get Accessible Organizations**
  (GET <host>/v1/orgs)
- **Get Patient List**
  (GET <host>/v1/orgs/{orgID}/patients)
- **Get Patient Details**
  (GET <host>/v1/patients/{ecn})

4.4 Instructions

1. Call the **Get Accessible Organizations** API to get a list of organizations you have access to (this may only be one organization ID).
2. With the organization ID you retrieved from step 1, call the **Get Patient List** API to get a list of ECNs in your organization.
   Note: We recommend that you save this list in your patient database. If you are set up as a patient level integrator, you will only receive ECNs that you have access to (see "3 Integrator types" on page 3).
3. Use the **Get Patient Details** API for each retrieved ECN to get the patient's first name, last name and date of birth.
4. Use the patient's first name, last name and date of birth to find a match in your system.
5. Once you find a match, associate the patient's ECN with your system's unique ID.

5 Create or update patients

5.1 Workflow overview

The instructions below outline how you can create or update a patient in your system with AVX.

Follow this workflow if your integrating application creates patients directly in AirView or if you need to add multiple physicians to a patient's record.

Note: The create and update APIs only allow for one physician to be defined per call. However, you can add a physician to the patient's record with each subsequent call.

5.2 Workflow tips

- The Patient ID field in AirView, which is different from the ECN, is an optional field that you can add when you create or update a patient record in AirView.
- You can only retrieve clinicians for specific locations. To get a list of all clinicians, you can go through each location systematically.
- In special-use cases, you should cache the list of organizations and locations.
  For example, when you configure an available subset per user or relatively static organization structure. If you cache, your application must refresh its cache when the organization structure changes because of an administrative action.
- If you want to update the patient's demographic information, patientId, setupDate or prescription details, you should not call to get the organization, location and clinical users. You can use the saved ECNs from your system to update the patient's demographic information. Only use these calls to update the patient's location or clinical user.

5.3 Required APIs

- **Get Accessible Organizations**
  (GET <host>/v1/orgs)
- **Get Locations for Organization**
  (GET <host>/v1/orgs/{orgID}/locations[?index={index}&count={count}])
- **Get Clinical Users**
  (GET <host>/v1/locations/{locationID}/clinicalUsers[?index={index}&count={count}])
- **Create Patient** (Create a new patient at a specific location for a clinical user)
  (POST <host>/v3/orgs/{orgID}/patients)

5.4 Instructions

1. Call the **Get Accessible Organizations** API to get a list of organizations you have access to (this may only be one organization ID).
2. For each organization, call the **Get Locations for Organization** API.
3. For each location returned, call the **Get Clinical Users** API to retrieve a list of clinical users in your organization.
   Note: We recommend that you save the location and clinical user lists in your integrating application and check for updates every two weeks.
4. Call the **Create Patient** API to create a patient and assign them to a location and clinical user.
   Note: Alternatively, you can update the patient at this step.

When you complete this procedure, the API returns an ECN for the newly created patient. We recommend that you store this ECN to access other AVX APIs in the future.

Note: When the API displays the locations and clinical users, call the **Get Locations for Organization** API only when you select a specific organization. Next, call the **Get Clinical Users** API only when you select a specific location from the list of organizations.

6 Get patient details

6.1 Workflow overview

The instructions below outline how you can retrieve a patient's details from AirView.

Follow this workflow to retrieve a patient's demographics and basic device information.

Note: You can only use this workflow if you store patient ECNs in your system. We recommend that you only use this API once every one to two weeks.

6.2 Workflow tips

- You can always find a patient's ECN at the end of the AirView URL.
  For example, https://<host>/patients/{ecn}
- The **Get Patient List** API does not return ECNs in any order.
- Brightree® IDs are not stored in AirView. You can only enter the Brightree ID in the AirView Patient ID field once you create a patient through the Brightree/AirView integration.
- If you want to keep your system's patient demographic details up to date with changes in AirView, we recommend that you call the **Get Patient Details** API once every two weeks or monthly as these details are not updated often.

6.3 Required APIs

- **Get Accessible Organizations**
  (GET <host>/v1/orgs)
- **Get Patient List**
  (GET <host>/v1/orgs/{orgID}/patients)
- **Get Patient Details**
  (GET <host>/v1/patients/{ecn})

6.4 Instructions

Use the stored ECNs from your database to call the **Get Patient Details** API.

7 Create a report or compliance list

7.1 Workflow overview

The instructions below outline how you can get a list of compliant patients and store each patient's compliance report in your system.

Follow this workflow to automatically get a list of compliant patients or to automatically download compliance reports.

7.2 Workflow tips

- We recommend that you call the PDF report APIs on demand. Do not use scheduled batch processing for PDF reports.
- If a patient becomes compliant, AirView creates a compliance met date. The **Get Compliant Patients for an Organization** API returns this date.
- The dateMetCompliance does not always fall within the range defined by the parameters (noOfDays and endRecievedOnDate). The compliant patients that appear on that list are based on the upload date of the data that made the patient compliant according to compliance rules.
  For example, there is a card-only patient who uses their device for more than four hours daily with a low AHI and mask leak. They have a follow-up appointment with their physician after 90 days on therapy to upload the data and go over the results. The compliance report shows that they were compliant after the first 30 days. For the patient to be on the list of patients who met compliance, the endReceivedOnDate and noOfDays need to include the day that they met with their physician to upload the data (day 90) rather than the day that they became compliant according to compliant usage rules (day 30).

7.3 Required APIs

- **Get Accessible Organizations**
  (GET <host>/v1/orgs)
- **Get Compliant List**
  (GET <host>/v1/orgs/{orgID}/compliantPatients?noOfDays={value}
  &endReceivedOnDate={date}[&index={index}&count={count}])
- **Get Compliance Report PDF**
  (GET <host>/v1/patients/{ecn}/reports/compliance?periodType=
  {periodType}&endDate={endDate}&noOfDays={days}&language={language})

7.4 Instructions

Note: Start on step 2 if you already have the organization ID.

1. Call the **Get Accessible Organizations** API to get a list of organizations you have access to (this may only be one organization ID).
2. Use the organization ID from step 1 and the numberOfDays and endReceivedOn parameters to call the **Get Compliant List** API.
   The API returns a list of compliant patients and the date each patient met initial compliance.
3. Use the ECN retrieved from step 2 to mark the patients as compliant (initial compliance only) in your system.
4. (Optional) If you want the compliance report PDF, call the **Get Compliance Report PDF** API for each ECN you retrieved in step 2.
   Note: Store this PDF in your system for easy access.

ResMed

ResMed Ltd
1 Elizabeth Macarthur Drive Bella Vista NSW 2153 Australia

 2281470/2 2020-04

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